

2025 FSAE-B: EV TECHNICAL INSPECTION

IMPORTANT: Each page must stay with the car until that specific part of inspection has been completed.  
Present the vehicle for inspection in the following order:

- 1a. Supporting Equipment Inspection
- 1b. Mechanical Technical Inspection
- 2. Rain Test
- 3. Weighing
- 3. Tilt Table
- 4. Brakes

|                        |  |           |
|------------------------|--|-----------|
| CAR NUMBER             |  | Team Name |
| SCHOOL NAME            |  |           |
| Captain name           |  |           |
| Cell number (Whatsapp) |  |           |

NOTE: If there is a conflict between this form and the rules, the rules prevail

| Check from the Technical Inspection Secretariat |       |                   |       |       |            |       |       |            |       |       |            |
|-------------------------------------------------|-------|-------------------|-------|-------|------------|-------|-------|------------|-------|-------|------------|
| Date:                                           | Hour: | Garage Inspection | Date: | Hour: | Chassis    | Date: | Hour: | Body       | Date: | Hour: | EV         |
|                                                 |       | Judge Name        |       |       | Judge Name |       |       | Judge Name |       |       | Judge Name |
| Date:                                           | Hour: | "5" Seconds       | Date: | Hour: | Rain Test  | Date: | Hour: | Weighing   | Date: | Hour: | Tilt Table |
|                                                 |       | Judge Name        |       |       | Judge Name |       |       | Judge Name |       |       | Judge Name |
| Date:                                           | Hour: | Brake Test        |       |       |            |       |       |            |       |       |            |
|                                                 |       | Judge Name        |       |       |            |       |       |            |       |       |            |

Technical Inspection Committee Final Approval  
Name:

or Senior Judge approval

|  |  |
|--|--|
|  |  |
|--|--|

Finished Date:

Finished Hour:

Comments:

|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |       |         |            |       |  |                                          |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------|------------|-------|--|------------------------------------------|
| <b>1 - Garage Inspection</b><br><b>Supporting Equipment Inspection: IN.3</b><br><div> <div>Check date</div> <div></div> </div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                | Name | Judge | Judge   | Start time | CAR # |  | 1A                                       |
|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |       |         | End time   |       |  |                                          |
|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |       |         |            |       |  |                                          |
|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |       |         |            |       |  | Recheck (Write on the back if necessary) |
| Item                                                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                    | Team | Check | Recheck |            |       |  |                                          |
| 1                                                                                                                              | <b>HELMETS - VE.3.2</b><br>- Closed-face with integral chin guard (no dirtbike helmets).<br>- Face shield integral with helmet, impact resistant material.<br>- Specification: Snell: KSnell: K2010, K2015, K2020, M2010, M2015, M2020-D,-R, M2025-D,-R, SA2010, SA2015, SA2020, SA2025, EA2016; or SFI: 31.1/2015; 41.1/2015; or FIA: 8860-2010 & newer, 8859-2015 & newer.<br>- No camera mounts: VE.2.5.3<br>- (Place sticker on left side) |      |       |         |            |       |  |                                          |
| 2                                                                                                                              | <b>BALACLAVA - VE.3.3.3</b><br>- Required for all drivers. Flame Resistant material. Covers head, neck and hair. (No label required.)                                                                                                                                                                                                                                                                                                          |      |       |         |            |       |  |                                          |
| 3                                                                                                                              | <b>ARM RESTRAINTS - VE.3.3.7</b><br>- Required for all drivers. Must be commercially manufactured. (No label required.)                                                                                                                                                                                                                                                                                                                        |      |       |         |            |       |  |                                          |
| 4                                                                                                                              | <b>DRIVERS' SUITS - VE.3.3.1</b><br>- Single piece suit, no holes. Must be labeled. Specification: SFI 3.2A/5; SFI 3.4/5; FIA 1986; FIA 8856-2000 or 8856-2018                                                                                                                                                                                                                                                                                 |      |       |         |            |       |  |                                          |
| 5                                                                                                                              | <b>GLOVES - VE.3.3.6</b><br>- Flame Resistant material (not all-leather). No holes. Leather palms allowed only over flame resistant material. (No label required.)                                                                                                                                                                                                                                                                             |      |       |         |            |       |  |                                          |
| 6                                                                                                                              | <b>SHOES - VE.3.3.5</b><br>No holes. Must be labeled. Specification: SFI 3.3; or FIA 8856-2000 or 8856-2018                                                                                                                                                                                                                                                                                                                                    |      |       |         |            |       |  |                                          |
| 7                                                                                                                              | <b>SOCKS - VE.3.3.4</b><br>Flame resistant material (no cotton; no polyester). Must cover all bare skin. (No label required.) <b>VE.3.1.2 Flame Resistant Material</b> : For this section some, but not all, of the approved materials are: Carbon X, Indura, Nomex, Polybenzimidazole (common name PBI) and Proban                                                                                                                            |      |       |         |            |       |  |                                          |
| 8                                                                                                                              | <b>WET TIRES - V.4.3.2</b> 2.4mm min tread depth. Molded or cut by tire manufacturer.<br><div> <div>Front</div> <div>Rear</div> <div>Brand:</div> <div>Brand:</div> <div>Compound:</div> <div>Compound:</div> <div>Size</div> <div>Size</div> </div>                                                                                                                                                                                           |      |       |         |            |       |  |                                          |
| 9                                                                                                                              | <b>FIRE EXTINGUISHERS - VE.2.3</b><br>•Hand-held, dry chemical (no AFFF or halon), min specification: 10BC; 1A10BC; 34B; 5A 34B; 20BE; or 1A 10BE.<br>•Min 0.9 kg (2 lb). •Two required: must present both at Tech.                                                                                                                                                                                                                            |      |       |         |            |       |  |                                          |

Comments

## 2 - Body Inspection

|            |  |
|------------|--|
| Check date |  |
|------------|--|

## Driver In, Bodywork On

| Item | Description | Team | Check | Recheck | Recheck (Write on the back if necessary) |
|------|-------------|------|-------|---------|------------------------------------------|
|------|-------------|------|-------|---------|------------------------------------------|

**Reminder to teams: bring the following items:**

- Car on “Dry Tires”  
 •Monocoque Laminate Test Specimens (if applicable) / •The Impact Attenuator that was tested (if applicable)  
 •Accumulator container and Tractive System firewall samples (if applicable) / •Electronic, or printed, copies of any submitted Rules Questions including responses  
 For egress (1 driver): •Helmet •Gloves •Arm Restraints •Long pants •Closed-toe shoes

|    |                                                                                                                                                                                                                                         |  |  |  |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
| 10 | <b>WET TIRES - V.4.3.1</b><br>Brand:                      Compound:                      Size:                                                                                                                                          |  |  |  |  |
| 11 | <b>First Year Vehicle - GR.9.3</b><br>New Chassis and is within its initial Competition Year.                                                                                                                                           |  |  |  |  |
| 12 | <b>PUSH BAR - VE.2.2</b><br>Always with car (detachable), push & pull, usable by 2 people standing behind the car. EVs: HV disconnect tool, if used.                                                                                    |  |  |  |  |
| 16 | <b>CAR NUMBERS - VE.1.1</b><br>On •front & both sides of car, •minimum 150 mm tall, •18 mm stroke & spacing. •White-on-black or B-on-W only. •Background shape: round, oval, rectangular or square. •Visibility must not be obstructed. |  |  |  |  |
| 17 | <b>SAE DECALS - VE.1.3</b> SAE logo on front OR both sides, in prominent location                                                                                                                                                       |  |  |  |  |
| 18 | <b>SCHOOL NAME - VE.1.2</b><br>School name or recognized initials. •Both sides of car, •easily visible location, •50 mm tall min, •Roman letters, •high contrast background.                                                            |  |  |  |  |
| 19 | <b>TECH STICKER SPACE - VE.1.4</b><br>25 cm wide x 20 cm high available space, located on centerline of upper front bodywork (nose) of car.                                                                                             |  |  |  |  |
| 20 | <b>CAMERA MOUNTS - VE.2.5</b><br>•If >0.25 kg, must be secured by two points (typical GoPro-brand camera is <0.25 kg). •No cameras mounted to helmet.                                                                                   |  |  |  |  |
| 21 | <b>GROUND CLEARANCE - V.1.4</b><br>Sufficient clearance so that no part of the car other than the tires will contact the track surface.                                                                                                 |  |  |  |  |
| 23 | <b>VISIBILITY - V.2.2</b><br>100° min field of view to each side. Head rotation OK, or mirrors. If mirrors, must be firmly installed and adjusted                                                                                       |  |  |  |  |
| 24 | <b>ROLL BAR PADDING - T.2.9</b><br>Installed on any bar that could be hit by the driver's helmet. •12 mm thick.<br>•Specification: SFI 45.1 or FIA 8857/2001 (pipe insulation or other foams not OK).                                   |  |  |  |  |
| 25 | <b>OTHER SIDE TUBES - F.5.12</b><br>Cockpit design must prevent driver's neck from hitting bracing or other side tubes                                                                                                                  |  |  |  |  |
| 26 | <b>VEHICLE CONTROLS - T.1.4</b><br>No hands, arms, or elbows outside side impact system when actuating controls. All controls, including shifter, must be inside cockpit and below topmost point of Front Hoop.                         |  |  |  |  |
| 27 | <b>LAP BELT FIT - T.2.5</b><br>•Must pass over pelvis, not waist. •45-65° to horizon for upright driver, 60-80° for reclined                                                                                                            |  |  |  |  |
| 28 | <b>SHOULDER HARNESS FIT - T.2.6</b><br>Angle from shoulder between 10° up and 20° down (vs horiz).                                                                                                                                      |  |  |  |  |
| 29 | <b>SUB BELT FIT - T.2.7</b><br>•Snug, holding latch in place. •Position in side-view: 5 Point: aligned with or forward of shoulder belt line; 6 Point: vertical or rearward of latch.                                                   |  |  |  |  |
| 30 | <b>ARM RESTRAINTS FIT - VE.3.3.7</b><br>Installed so the driver can release them and exit unassisted regardless of vehicle's position.                                                                                                  |  |  |  |  |

| 2 - Body Inspection |                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Name                                                                                                                                   | Judge | Judge   | EV Vehicle Number                        |  | 2B |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------|---------|------------------------------------------|--|----|
| Check date          |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                        |       |         |                                          |  |    |
| Driver Fit          |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                        |       |         |                                          |  |    |
| Item                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               | Team                                                                                                                                   | Check | Recheck | Recheck (Write on the back if necessary) |  |    |
| 31                  | <b>HEAD RESTRAINT FIT - T.2.8</b><br>•Max 25 mm (1") forward gap to helmet. •Helmet contact point min 50 mm (2") from any edge. <b>APPLIES TO ALL DRIVERS (may be adjusted for each driver)</b>                                                                                                                                                                                                                                                           |                                                                                                                                        |       |         |                                          |  |    |
| 32                  | <b>MAIN HOOP &amp; FRONT HOOP HEIGHTS - F.5.6.3</b><br>Helmet 50 mm (2.0") below lines between •top of front and main roll hoops, and •top of main hoop to rear attachment point of main hoop bracing. <b>(Applies for every driver.)</b>                                                                                                                                                                                                                 |                                                                                                                                        |       |         |                                          |  |    |
| 33                  | <b>DRIVER'S FOOT PROTECTION - T.1.3.2</b><br>Feet must be rearward of the Front Bulkhead and no part of shoes or legs above or outside the Major Structure in side or front views when touching pedals. Remove nose bodywork if necessary for visual access                                                                                                                                                                                               |                                                                                                                                        |       |         |                                          |  |    |
| 34                  | <b>EGRESS - IN.5.2</b> 5 seconds max to actuate cockpit master switch and exit to side of vehicle, from driving position, wearing safety equipment. Wings must remain fixed in position. (See egress worksheet.)                                                                                                                                                                                                                                          | <b>Attention : It is not necessary to analyze at the technical inspection site, 02 exclusive boxes for the 5-second check / Egress</b> |       |         |                                          |  |    |
| Driver Out          |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                        |       |         |                                          |  |    |
| 35                  | <b>BODY &amp; STYLING - V.1</b><br>Open cockpit, formula style body. Four wheels, not in a line.                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                        |       |         |                                          |  |    |
| 36                  | <b>OPEN WHEEL - V.1.1</b><br>•Top 180° of wheel/tire unobstructed from above. •Tires unobstructed from sides. •Vertical keepout zones 75mm in front & behind tires.                                                                                                                                                                                                                                                                                       |                                                                                                                                        |       |         |                                          |  |    |
| 37                  | <b>BODYWORK - T.7.2</b><br>•No large openings (> ~6mm) in bodywork into driver compartment in front of or alongside driver.<br>•Body/nose min 38 mm radius, +/-45° all directions.                                                                                                                                                                                                                                                                        |                                                                                                                                        |       |         |                                          |  |    |
| 38                  | <b>WHEELBASE - V.1.2</b><br>Minimum 1525 mm.                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                        |       |         |                                          |  |    |
| 39                  | <b>JACKING POINT - VE.2.1</b><br>•Horizontal, lateral tube at the rear. •Orange color. •300 mm wide by 25-30 mm O.D. •Visible to person standing 1 meter behind car. •75 mm min ground clearance •Rear tires must come off the ground using Quick-Jack (lifts to 200 mm). Exposed around the lower 180° of its circumference                                                                                                                              |                                                                                                                                        |       |         |                                          |  |    |
| 40                  | <b>SEAT - T.1.5, T.1.6</b><br>•Insulated against heat conduction, convection and radiation.<br>•Lowest point <b>no lower than top of lower structure</b> OR must have longitudinal 1.00" OD x 0.065" steel tube underneath.                                                                                                                                                                                                                               |                                                                                                                                        |       |         |                                          |  |    |
| 41                  | <b>FIREWALL - T.1.8</b><br>Rigid, nonflammable material. •Separates driver (line-of-sight to mid-height of driver's helmet) from fuel, cooling, oil, lithium battery, heat sources, and all tractive system components. Wire/cable pass-throughs OK with grommets. Multiple panels OK w/ gaps sealed. • <b>Seals (e.g.: metal tape) cannot be stressed or used to hold firewalls in place.</b> •No gaps at sides or bottom. •Leaks cannot contact driver. |                                                                                                                                        |       |         |                                          |  |    |
| 42                  | <b>FLOOR CLOSEOUT PANEL - T.1.7</b><br>Non-perforated, non-brittle material from foot area to firewall. Multiple panels OK if gaps less than 3 mm (1/8").                                                                                                                                                                                                                                                                                                 |                                                                                                                                        |       |         |                                          |  |    |
| Driver Out          |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                        |       |         |                                          |  |    |
| 43                  | <b>DRIVER'S LEG PROTECTION - T.1.3</b><br>Covers inside cockpit over sharp parts or moving suspension and steering components.                                                                                                                                                                                                                                                                                                                            |                                                                                                                                        |       |         |                                          |  |    |
| 48                  | <b>WHEELS - V.4</b><br>•203 mm (8.0") min diameter. •Wheels with single wheel nut must have positive retainer. •Aluminum lug nuts hard-anodized and pristine condition.                                                                                                                                                                                                                                                                                   |                                                                                                                                        |       |         |                                          |  |    |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------|------------------------------------------|--|----|
| 2 - Body Inspection |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Name | Judge | Judge   | EV Vehicle Number                        |  | 2C |
|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
| Check date          |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
| Driver Restraints   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
| Item                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Team | Check | Recheck | Recheck (Write on the back if necessary) |  |    |
| 49                  | <b>DRIVER RESTRAINT HARNESS - T.2.2</b><br>5, 6 or 7 point and be •labeled: SFI 16.1, 16.5, or FIA 8853/98, 8853/2016 (or lookup FIA D-number) •All lap belts must have Quick Adjusters. •Reclined drivers must have 6 or 7 point, and Quick Adjuster sub-belts OR 2 sets of sub belts. •Belts expire after expiration year (not month)marked on label (SFI & FIA).                                                                                          |      |       |         |                                          |  |    |
| 50                  | <b>HARNESS HARDWARE AND INSTALLATION - T.2.2</b><br>•Belts threaded through hardware per mfr instructions. •Hardware must be unmodified (no drilling, welding, etc).                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| 51                  | <b>HARNESS MOUNTS - T.2.4</b><br>•Belts must be protected by firewalls. •All belts attached securely to Primary Structure. •Tabs 1.6 mm (0.063”) thick min, 60 mm2 shear area; tabs combining lap & sub belts 90 mm2 . Double-shear preferred. •Tabs welded on both sides; bolt-on tabs use minimum of two ¼” dia Grade 5 bolts. •Tabs aligned with load direction of belt, may be pre-bent.                                                                 |      |       |         |                                          |  |    |
| 52                  | <b>LAP BELT POSITION - T.2.5</b><br>•Pivoting mounting using eye bolt or shoulder bolt (no tube wrap). •Not re-directed by seat. •Belt centerline max 3” forward of seatback-seatbottom junction in side view                                                                                                                                                                                                                                                |      |       |         |                                          |  |    |
| 53                  | <b>SHOULDER HARNESS POSITION - T.2.6</b><br>Mounting points 175-235 mm (~7”- 9”) apart. Tube wrap OK                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| 54                  | <b>SUB BELT POSITION - T.2.7</b><br>Sub belts cannot be re-directed by anything other than the driver's body.                                                                                                                                                                                                                                                                                                                                                |      |       |         |                                          |  |    |
| 55                  | <b>BELT ATTACHMENT FASTENERS - T.2.5.8, T.2.6.3, T.2.7.4</b><br>•Lap & Shoulder 10 mm Grade 8.8 (3/8” SAE Grade 5), •Sub 8 mm (5/16”), or as specified by harness manufacturer. •Pins required in clip-brackets •Special rules for eye-type attachments: aligned, fully tight, see T.2.4.5 •CRITICAL (positive locking) - T.8                                                                                                                                |      |       |         |                                          |  |    |
| 56                  | <b>HEAD RESTRAINT - T.2.8</b><br>•Min 150 mm width •235 cm2 area AND height adjustment of 175 mm (7”); OR 280 mm (11”) height. •38 mm (1.5”) thick. •Near vertical. •Pad and mount must take 900 N rearward and 300 N lateral forces. •Energy absorbing material: SFI Standard 45.2 or FIA Tech List 17 (CONFOR blue CF45 or CF45M), with 20mm inspection hole in its cover. •Mount must be within Main Hoop Bracing envelope, or add extra tube per F.5.10. |      |       |         |                                          |  |    |
| Aero                |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
| S33                 | <b>AERODYNAMIC DEVICES - T.7.1</b><br>•No powered ground effects. •Securely mounted - no oscillation or excessive movement. (Wings, undertray, splitter, endplates, vanes, etc.) •Vertical keepout zones 75mm in front & behind tires.                                                                                                                                                                                                                       |      |       |         |                                          |  |    |
| S34                 | <b>EDGE RADII - T.7.1</b><br>•Horizontal leading edges min 5 mm radius. •Vertical forward-facing edges min 3 mm radius. •Other edges: not sharp - GR.1.5                                                                                                                                                                                                                                                                                                     |      |       |         |                                          |  |    |
| S35                 | <b>FRONT MOUNTED AERO - T.7.4</b><br>•Max forward: 700 mm ahead of the front tires. •Max width: OUTSIDE of the front tires (at hub height). •Max height: 250 mm if in front of tires (w/out driver).                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| S36                 | <b>REAR MOUNTED AERO - T.7.5-8</b><br>•Max rearward: 250 mm behind the rear tires. •Max forward: Headrest support (undertrays exempt). •Max width: INSIDE of the rear tires (at hub height). •Max height: 1.2 m above ground (w/out driver).                                                                                                                                                                                                                 |      |       |         |                                          |  |    |
| S37                 | <b>AERO BETWEEN WHEEL CENTERLINES - T.7.5-8</b><br>•Max width: Line between front & rear tires (at hub height). •Max height: 500 mm (exempt if w/in 400 mm of car centerline).                                                                                                                                                                                                                                                                               |      |       |         |                                          |  |    |
| Comments            |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |

|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         |                                          |       |  |    |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------|------------------------------------------|-------|--|----|
| 3 - Chassis Inspection                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                        | Name | Judge | Judge   | Start time                               | CAR # |  | 3A |
|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         |                                          |       |  |    |
|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         | End time                                 |       |  |    |
| Check date                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         |                                          |       |  |    |
| Bodywork Off                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         |                                          |       |  |    |
| Item                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                            | Team | Check | Recheck | Recheck (Write on the back if necessary) |       |  |    |
| 44                                                                                     | NON-CRUSHABLE OBJECTS - F.8.6.3<br>25 mm clearance aft of AI Plate, <i>Front Bulkhead including diagonals, and to pedals @ full travel and adjustment.</i>                                                                                                                                                                                                                                                                             |      |       |         |                                          |       |  |    |
| 46                                                                                     | BRAKE PEDAL - T.3.2<br>Steel, aluminum, or machined titanium (no welded Ti). <b>Alt matl. OK for pad face.</b><br>Entire pedal-to-frame system capable of 2kN (tested only by organizers).                                                                                                                                                                                                                                             |      |       |         |                                          |       |  |    |
| General Inspection - Chassis / Check Chassis with the SES Chassis FSAE-B 2024 document |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         |                                          |       |  |    |
| 57                                                                                     | STEERING - V.3.2<br>•All steerable wheels must have positive stops to prevent linkage lock-up or contact with other parts. •7° max freeplay at the steering wheel. •NO STEER-BY-WIRE on front wheels. •No cables or belts. •No bonded joints in column without metal backup. •Rear steer limited to 6° total, with mechanical stops.                                                                                                   |      |       |         |                                          |       |  |    |
| 58                                                                                     | CRITICAL FASTENERS, Steering - V.3.2.8<br>Steering wheel, column, rack mounting, tie rods                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |       |  |    |
| 59                                                                                     | SUSPENSION PICK-UP POINTS - GR.1.5<br>Inspected thoroughly for integrity: binding, over-articulation.                                                                                                                                                                                                                                                                                                                                  |      |       |         |                                          |       |  |    |
| 60                                                                                     | STEERING WHEEL - V.3.3<br>•Continuous perimeter, near round (no concave sections). •Driver operable quick disconnect. •Not higher than top of Front Hoop, in any angular position. •250 mm max rearward of Front Hoop (F.5.7.5).                                                                                                                                                                                                       |      |       |         |                                          |       |  |    |
| 61                                                                                     | SUSPENSION - V.3.1<br>•Full suspension including front and rear damping. •Spherical rod ends and bearings: double-shear or safety washers.                                                                                                                                                                                                                                                                                             |      |       |         |                                          |       |  |    |
| 62                                                                                     | MODIFIED LUG BOLTS/STUDS - V.4.2.2<br>Verify good engineering practices are followed e.g. no drill holes for mass reduction.                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |       |  |    |
| 63                                                                                     | CRITICAL FASTENERS: SUSPENSION - V.3.1.4<br>Control arms, knuckle, spring load path, single wheel nuts. Exempt: lug nuts, multi-piece wheels, anti-roll bars, dampers. All fasteners must be tight (esp: jam nuts).                                                                                                                                                                                                                    |      |       |         |                                          |       |  |    |
| 64                                                                                     | BRAKES - T.3.1<br>•Single pedal actuates all 4 wheels (one brake on limited slip OK) •Two separate hydraulic circuits w/ reservoirs; no brake-by-wire. •Protected by structure/shields from drivetrain & collisions. •No plastic brake lines. •No parts below chassis/tub in side-view.                                                                                                                                                |      |       |         |                                          |       |  |    |
| 65                                                                                     | CRITICAL FASTENERS: BRAKES - T.3.1<br>•Pedal Assembly: including adjustment mechanism, caliper to knuckle mounts, rotor to hat. •Exempt: COTS caliper body assembly                                                                                                                                                                                                                                                                    |      |       |         |                                          |       |  |    |
| 66                                                                                     | BRAKE CONTROL SYSTEMS - GR.1.5<br>ABS, Traction, Yaw Control, etc: must have an approved FMEA.                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |       |  |    |
| Attention                                                                              | Critical Fasteners Reference: T.8<br>For all locations marked "CRITICAL FASTENERS" in this form, the following requirements apply: •Material rating: SAE Gr 5, Metric 8.8, or AN/MS. •Hex-head or socket-head; no pan head, button head, flat head, or low-profile screws. •Positive locking: safety wire, lock pins, or locking nuts with 2 threads visible beyond nut. Threadlock compounds not sufficient; no nylock nuts if >80 °C |      |       |         |                                          |       |  |    |
| Comments                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |         |                                          |       |  |    |

|                        |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------|---------|------------------------------------------|--|----|
| 3 - Chassis Inspection |                                                                                                                                                                                                                                                                                                                                                                                               | Name                            | Judge | Judge   | EV Vehicle Number                        |  | 3B |
|                        |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |
| Check date             |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |
| Roll Hoops             |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |
| Item                   | Description                                                                                                                                                                                                                                                                                                                                                                                   | Team                            | Check | Recheck | Recheck (Write on the back if necessary) |  |    |
| S1                     | MAIN HOOP - F.5.8<br>•Must be steel with smooth bends with no wrinkles. •Must be 1 piece & extend to lowest frame member. •Tube endpoints 380 mm apart (inside dim) at bottom attachment. •Above Upper SIS, max 10° to vertical. •Bottom ends, no more than 10° rearward from vertical.                                                                                                       |                                 |       |         |                                          |  |    |
| S2                     | MAIN HOOP BRACING - F.5.9, F.6.6<br>Must be steel w/ no bends. •One brace each side, attached within 160 mm of top of hoop. •Min 30° included angle with hoop. •Bracing must not be on same side of vertical as Main Hoop.                                                                                                                                                                    |                                 |       |         |                                          |  |    |
| S3                     | HEAD RESTRAINT - T.2.8.3<br>•Inside Rollover Envelope or F.5.10 Head Restraint protection.                                                                                                                                                                                                                                                                                                    |                                 |       |         |                                          |  |    |
| S4                     | EXTERNAL ITEMS - F.5.11 (e.g. wings, suspension mounts, radiator) Single fastener connection at MH-MHB node preferred. For mid-MHB or mid-Upper-MH mounts: SES failure calculations and structural brace required. Some light items may freely attach per F.5.11.3. Components offset longitudinally, not resting on tube. Suspension only permitted at nodes. Powertrain mounts discouraged. |                                 |       |         |                                          |  |    |
| S5                     | FRONT HOOP - F.5.7<br>•Closed-section metal tube (may be multi-piece). •Non-ferrous tubes must have inspection holes. •Must extend to lowest frame member. •Above Upper SIS, max 20° to vertical, F.6.3.5 max 10° towards cockpit if no rearward bracing.                                                                                                                                     |                                 |       |         |                                          |  |    |
| S5                     | BENDS IN ROLL HOOPS – F5.6.3<br>•Front- and side-view bends below the respective Roll Hoop Braces must be: triangulated in side-view OR less than 25mm from a monocoque attachment point..                                                                                                                                                                                                    |                                 |       |         |                                          |  |    |
| Chassis Features       |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |
| S7                     | STEERING COMPONENTS OUTSIDE FRAME - F.5.14<br>•Rack inside primary structure. •F.5.14 Steering rack outside primary structure protected to top, bottom, and local chassis width. •Welded to chassis, or 30kN per end (60kN total) shown in SES.                                                                                                                                               |                                 |       |         |                                          |  |    |
| S9                     | HOLES AND OPENINGS IN REGULATED TUBES - F.5.3<br>•Holes in regulated tubes addressed in SES. •Tube ends must be closed using welded caps or welded inserts, except bottom of roll hoops.                                                                                                                                                                                                      |                                 |       |         |                                          |  |    |
| S10                    | BOLTED JOINTS CONNECTING PRIMARY STRUCTURES F.5.4, F.5.13<br>•Double-shear (capped) or Sleeved butt joints only. •No bends. •No rod-end bearings. •Edge of any bolt hole located > 1.5 x hole diameter from nearest edge of the material. •No blind or welded threaded fasteners. •Bolts 8 mm (5/16"), plates 2.0 mm (0.08").                                                                 |                                 |       |         |                                          |  |    |
| S11                    | HARNES ATTACHMENTS •T.2.4.3<br>•T.2.4.3 Tube: 15kN, single shear per SES. •F.7.9 Composite: Test specimens, representative of vehicle. •Test loading direction: prefer normal, min 45 up from panel or harness installation per rules. •Tabs 1.6 mm (0.063") thick min, meets tear-out strength requirements in SES. Double-shear preferred.                                                  |                                 |       |         |                                          |  |    |
| Front Impact           |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |
| S12                    | IA LEADING EDGE VERTICAL POSITION - F.8.5.6<br>•Bottom leading edge of IA must be lower than 150mm above lowest point of top surface of lower SIS tube. •Honeycomb must be horizontal; foam may be vertical. • V.1.4.2 Ride Height below lowest point of Lower SIS should be <=75mm, and must be <=90mm.                                                                                      | Record ride height measurement: |       |         | Record ride height measurement:          |  |    |
| S13                    | STANDARD IMPACT ATTENUATOR - F.8.4.3<br>•Must be fully bonded to AIP. •IA must not be degraded or damaged.                                                                                                                                                                                                                                                                                    |                                 |       |         |                                          |  |    |
| Commnets               |                                                                                                                                                                                                                                                                                                                                                                                               |                                 |       |         |                                          |  |    |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
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| 3 - Chassis Inspection   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Name           | Judge | Judge             | EV Vehicle Number                        |  | 3C |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| Check date               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| Front Impact (Continued) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| Item                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Team           | Check | Recheck           | Recheck (Write on the back if necessary) |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S14                      | NON-STANDARD IMPACT ATTENUATOR - F.8.4•200 mm long x 200 mm wide x 100 mm high. •Adhesive attachment required for foam or honeycomb. •Attached to AIP w/ welds - 1:1 weld ratio, beads 25 mm min, or eight 8 mm (5/16") bolts. •Must be capable of taking vertical and transverse loads. •Must be identical to test specimen, engraved dates of test. •Suitable failure mode (crushed element, not collapsed mount).                                                                                                                                                                                                                                                                                  |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S15                      | ANTI-INTRUSION PLATE - F.8.2 •1.5 mm steel or 4 mm aluminum. •120KN physical test, or 3-point composite per SES. •Attached to bulkhead w/ eight 8 mm (5/16") bolts, or bonded or laminated (AIP extending to outer edge of bulkhead), or >50% welded (AIP extending to centerline of bulkhead tubes). •Quick release and shear pin per SES if present. •CRITICAL FASTENERS: F.8.5.5                                                                                                                                                                                                                                                                                                                   |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S16                      | FRONT WING DETACHMENT - F.8.6<br>•No portion of front wing or mountings block the front view of bulkhead.<br>•F.8.6.2 Detaches below peak load. •Fasteners match SES documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S17                      | FRONT BULKHEAD DIAGONAL - F.8.4.3<br>•Unless tested for <25mm AIP deflection, diagonal structure is required for the standard foam IA when FBH > 400mm x 350mm; diagonal required for all standard honeycomb IAs. •Diagonal may be attached to back of AIP.                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| Tube Primary Structure   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S18                      | LOAD PATHS AND TRIANGULATION - F.1.17 F.1.17 •Upper and lower load paths, with sufficient diagonal tubing to fully triangulate in between, are required continuously through the FBHS, SIS, MHBS. •Minimum tube size requirements apply to Upper, Lower, and Diagonal structural tubes.                                                                                                                                                                                                                                                                                                                                                                                                               |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S19                      | BENT OR MULTIPLE TUBES - F.5.2<br>•Min bend radius at centerline: 3x tube OD. •A brace must connect from on the bend to a frame node, •Within 30° of plane of bend (Upper SIS, Shoulder Harness Bar exempt from 30°). •Brace tube matches min, except Roll Hoop bends Size C, Upper SIS diag Size B, Shoulder Harness Brace Size C.                                                                                                                                                                                                                                                                                                                                                                   |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S20                      | FRONT BULKHEAD SUPPORT - F.6.2<br>•Connect bottoms of Front Bulkhead and Front Hoop; •Connect within 50mm of top of bulkhead, 100 mm above and 50 mm below upper SIS tube •Triangulated brace to SIS if top tube does not connect near upper SIS.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S21                      | FRONT HOOP BRACING - F.6.3<br>•Two forward facing braces, attached within 50 mm of top of hoop, attached to Front Bulkhead or triangulated structure. •Rearward bracing if FH more than 10° rear above Upper SIS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S22                      | SIDE IMPACT STRUCTURE - F.6.4<br>•Top surface of upper tube must be between 265 mm and 320 mm above top surface of lower SIS tube at its lowest point (F.6.4.3). •Lower tube must connect endpoints of Main and Front hoops.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S23                      | SHOULDER HARNESS MOUNTING BAR - F.6.5<br>•Uncut, closed-section tube. •Attached to Main Hoop. •Braced to Main Hoop if bent: minimum 30° between brace and bar. •Nothing attached except for Shoulder Harnesses, driver equipment, headrest, and light bodywork including the firewall and or seat.                                                                                                                                                                                                                                                                                                                                                                                                    |                |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| S24                      | TUBE MINIMUMS - F.3.2 Tech may use ultrasound to measure wall thickness and/or ask that 4 mm dia holes be drilled ( See table F.3.2.1 Requirements by Application)<br><table><tr><td>SIZE D</td><td></td><td>Bent Upper SIS</td><td></td><td></td><td></td></tr><tr><td>SIZE A</td><td></td><td>MH</td><td></td><td>FH</td><td></td></tr><tr><td>SIZE B</td><td></td><td>MHBrace</td><td></td><td>Lap/Anti</td><td></td></tr><tr><td></td><td></td><td>FHBrace</td><td></td><td>FHBrace &gt; 10 Rear</td><td></td></tr><tr><td>SIZE C</td><td></td><td>IADiag</td><td></td><td>FBHS</td><td></td></tr><tr><td></td><td></td><td>SHBrace</td><td></td><td>External Steering</td><td></td></tr></table> | SIZE D         |       | Bent Upper SIS    |                                          |  |    | SIZE A |  | MH |  | FH |  | SIZE B |  | MHBrace |  | Lap/Anti |  |  |  | FHBrace |  | FHBrace > 10 Rear |  | SIZE C |  | IADiag |  | FBHS |  |  |  | SHBrace |  | External Steering |  |  |  |  |  |  |  |
| SIZE D                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Bent Upper SIS |       |                   |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| SIZE A                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MH             |       | FH                |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| SIZE B                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MHBrace        |       | Lap/Anti          |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FHBrace        |       | FHBrace > 10 Rear |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
| SIZE C                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | IADiag         |       | FBHS              |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SHBrace        |       | External Steering |                                          |  |    |        |  |    |  |    |  |        |  |         |  |          |  |  |  |         |  |                   |  |        |  |        |  |      |  |  |  |         |  |                   |  |  |  |  |  |  |  |



### 3 - Chassis Inspection

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| Check date |  |
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## EV Impact Protection

| 3 - Chassis Inspection |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Name       | Judge | Judge   | EV Vehicle Number                        |  | 3D |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|---------|------------------------------------------|--|----|
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Check date |       |         |                                          |  |    |
| EV Impact Protection   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |       |         |                                          |  |    |
| Item                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Team       | Check | Recheck | Recheck (Write on the back if necessary) |  |    |
| S33 EV                 | <b>SIDE IMPACT - ACCUMULATOR AND TRACTIVE SYSTEM - F.11.2.1</b><br>• An upper, lower, and full triangulating diagonals (or SES equiv.) are required. • Up to Upper SIS height or the top of the accumulator, whichever is lower. • Material size B beside Accumulator, Size C elsewhere. • 25mm gap around the accumulator should be present except at mount locations.                                                                                                                                                                     |            |       |         |                                          |  |    |
|                        | <b>REAR IMPACT - ACCUMULATOR AND TRACTIVE SYSTEM - F.11.2.2</b><br>• An upper, lower, and fully triangulating diagonal (or SES equiv.) are required. • A fully structural path (triangulated if tube) back to the Upper and Lower SIS is required. • Reaching at least 240mm above the lower SIS. • Material Size B if Acc is <100mm away, Size C if Acc is >=100mm away. • Differential Mounts or Plates used as SES equiv. must fully overlap the upper and lower tubes. • Non-crushable objects should have 25mm gap to the accumulator. |            |       |         |                                          |  |    |

### Comments regarding Tube Chassis

### 3 - Chassis Inspection

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| Check date |  |
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#### Monocoque Chassis

| Item | Description | Team | Check | Recheck | Recheck (Write on the back if necessary) |
|------|-------------|------|-------|---------|------------------------------------------|
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#### Cockpit Templates ( Applicable to Tube and Monocoque Chassis)

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
| S25       | <b>LAMINATE TEST SPECIMENS - F.4.3</b><br>•Three-point bending: 275 x 500 mm or 138 x 500 mm •Perimeter shear: 100 x 100 mm or 3-point •Lap joint •Engraved with date, specimen name.                                                                                                                                                                                                                                                                           |  |  |  |  |
| S26       | <b>BOLTED ROLL HOOP ATTACHMENTS - F.7.4, F.7.6</b><br>•Three points per side x 30 kN each, or two x 45 kN. •Within 25mm of bends below top of monocoque. •Boxed tube to plates and gussets ••Mounting plates on hoop min 2mm thick. •Backing plates per SES, receipts for non-steel. •Two 8 mm bolts per joint. •One 10mm bolt on centerline allowed for hoop braces. •No blind or threaded inserts. •Requires inserts or no gap between inner and outer skins. |  |  |  |  |
| S27       | <b>LAMINATED FRONT HOOP - F.7.4</b><br>•Front Hoop may be fully laminated instead of bolting. •Must not be attached only by adhesive, cores must fit tightly.                                                                                                                                                                                                                                                                                                   |  |  |  |  |
| S28       | <b>SIDE IMPACT PROTECTION - F.7.5</b><br>•Min 290mm height above bottom surface of chassis. •No side view openings between Front Hoop and Main Hoop.                                                                                                                                                                                                                                                                                                            |  |  |  |  |
| S29       | <b>LAMINATE THICKNESS VS SAMPLES AND SES - 4mm hole or external calipers. PANELS</b><br>•FB •FBDiag •FBHS •FHBrace •SIS •Lap/Anti •SH •MHBS ATTACHMENTS<br>•SteeringProtection •AIP •FH •Lap/Anti •SH •MH •MHB                                                                                                                                                                                                                                                  |  |  |  |  |
| S30       | <b>MAIN HOOP &amp; FRONT HOOP HEIGHTS - F.5.6.5</b><br>Helmet of 95th percentile male (PERCY) to be 50 mm below the lines between top of front and main roll hoops and between top of main hoop to rear attachment point of main hoop bracing. Center of bottom circle placed minimum 915 mm from pedals.                                                                                                                                                       |  |  |  |  |
| S31       | <b>COCKPIT OPENING - T.1.1</b><br>Template to pass from above cockpit to bottom of top SIS tube and less than 320mm from lowest point inside cockpit. Steering wheel & column, seat and padding can be removed; no removing firewall. Fore/aft translation of template OK.                                                                                                                                                                                      |  |  |  |  |
| S32       | <b>COCKPIT INTERNAL CROSS SECTION T.1.2</b><br>Template to pass from rearwards of the steering column to 100 mm rearwards of the pedals. Steering wheel may be removed; padding may be removed if removable with no tools & with driver in seat.                                                                                                                                                                                                                |  |  |  |  |
| Attention | <b>Critical Fasteners Reference: T.8</b><br>For all locations marked “CRITICAL FASTENERS” in this form, the following requirements apply: •Material rating: SAE Gr 5, Metric 8.8, or AN/MS. •Hex-head or socket-head; no pan head, button head, flat head, or low-profile screws. •Positive locking: safety wire, lock pins, or locking nuts with 2 threads visible beyond nut. Threadlock compounds not sufficient; no nylock nuts if >80 °C                   |  |  |  |  |

|                                      |
|--------------------------------------|
| Comments regarding Monocoque Chassis |
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| Name | Judge | Judge   | EV Vehicle Number                        |  | 3E |
|      |       |         |                                          |  |    |
| Team | Check | Recheck | Recheck (Write on the back if necessary) |  |    |

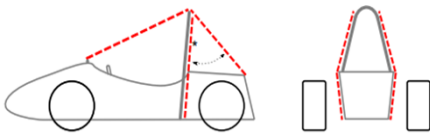
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| <b>4 - EV</b><br><div> <div>Check date</div> <div></div> </div>                             |                                                                                                                                                                                                                                                            | Name | Judge | Judge   | Start time                               | CAR #                                                                                 |  | 4A |
|                                                                                             |                                                                                                                                                                                                                                                            |      |       |         | End time                                 |                                                                                       |  |    |
|                                                                                             |                                                                                                                                                                                                                                                            |      |       |         |                                          |                                                                                       |  |    |
| EV Tractive System / Car Devices                                                            |                                                                                                                                                                                                                                                            |      |       |         |                                          |                                                                                       |  |    |
|                                                                                             | Name: (ESO must not be a driver, even if there's more than one)                                                                                                                                                                                            |      |       |         | Phone Number:                            |                                                                                       |  |    |
| ESO                                                                                         |                                                                                                                                                                                                                                                            |      |       |         |                                          |                                                                                       |  |    |
| ESA                                                                                         |                                                                                                                                                                                                                                                            |      |       |         |                                          |                                                                                       |  |    |
| Item                                                                                        | Description                                                                                                                                                                                                                                                | Team | Check | Recheck | Recheck (Write on the back if necessary) |                                                                                       |  |    |
| Personal Protective Equipment and External Checks / Start with accumulator container closed |                                                                                                                                                                                                                                                            |      |       |         |                                          |                                                                                       |  |    |
| EV1                                                                                         | <b>EV.2 / GR.7</b><br>Datasheets and copies of any submitted Rules Questions with the received answer.                                                                                                                                                     |      |       |         |                                          |                                                                                       |  |    |
| EV2                                                                                         | <b>IN.4.1 / BR.4.6.3.2</b><br>Safety glasses with side shields for all team members that might work on the Tractive System or Accumulator.                                                                                                                 |      |       |         |                                          |                                                                                       |  |    |
| EV3                                                                                         | <b>IN.4.1 / BR.4.6.3.2</b><br>Face Shield.                                                                                                                                                                                                                 |      |       |         |                                          |                                                                                       |  |    |
| EV4                                                                                         | <b>IN.4.1 / BR.4.6.3.2</b><br>HV insulating gloves which are 12 months or less from their test date. Must include leather gloves to protect the insulating gloves from mechanical damages. Test report must be provided.                                   |      |       |         |                                          |                                                                                       |  |    |
| EV5                                                                                         | <b>IN.4.1 / BR.4.6.3.2</b><br>Multimeter with protected probe tips.                                                                                                                                                                                        |      |       |         |                                          |                                                                                       |  |    |
| EV6                                                                                         | <b>IN.4.1 / BR.4.6.3.2</b><br>Insulated cable shears, screw drivers and all other tools used in TS.                                                                                                                                                        |      |       |         |                                          |                                                                                       |  |    |
| EV7                                                                                         | <b>IN.4.1 / BR.4.6.3.2</b><br>Two HV insulating blankets of minimum 0.83 [m²] each. Test report must be provided.                                                                                                                                          |      |       |         |                                          |                                                                                       |  |    |
| EV8                                                                                         | <b>EV.8.1</b><br>Charger(s) for the Accumulator(s).                                                                                                                                                                                                        |      |       |         |                                          |                                                                                       |  |    |
| EV9                                                                                         | <b>EV.4.10</b><br>Accumulator Container Hand Cart.                                                                                                                                                                                                         |      |       |         |                                          |                                                                                       |  |    |
| EV10                                                                                        | <b>EV.5.1.4</b><br>Spare Accumulator(s) (if applicable)                                                                                                                                                                                                    |      |       |         |                                          |                                                                                       |  |    |
| EV11                                                                                        | <b>BR.4.6.3.1</b><br>Two 4 mm banana-banana type measuring tips.                                                                                                                                                                                           |      |       |         |                                          |                                                                                       |  |    |
| EV12                                                                                        | <b>EV.4.3.8.a</b><br>School Name and Vehicle Number must be present and clearly visible on the accumulator.                                                                                                                                                |      |       |         |                                          |                                                                                       |  |    |
| EV13                                                                                        | <b>EV.4.3.8.b</b><br>Accumulator must have symbol specified in ISO 7010-W012 (triangle with black lightning bolt on yellow background).<br>•Triangle side length of 100mm minimum<br>•Visibility from all angles, including when the lid/cover is removed. |      |       |         | Example:                                 |  |  |    |
| EV14                                                                                        | <b>EV.4.3.8.c</b><br>"Always energized" text must be present and clearly visible on the accumulator.                                                                                                                                                       |      |       |         |                                          |                                                                                       |  |    |
| EV15                                                                                        | <b>EV.4.3.8.d</b><br>"High Voltage" text must be present and clear visible on the accumulator, if accumulator voltage is greater than 60VDC.                                                                                                               |      |       |         |                                          |                                                                                       |  |    |
| EV16                                                                                        | <b>EV.5.7.4.b</b><br>Voltage Indicator must be labeled with "High Voltage Present".                                                                                                                                                                        |      |       |         |                                          |                                                                                       |  |    |
| EV17                                                                                        | <b>F.10.4.4</b><br>Accumulator may have holes/openings with a maximum diameter of 10mm. Only round holes are allowed. Must not have a line of sight to the driver, with the firewall installed or not.                                                     |      |       |         |                                          |                                                                                       |  |    |
| EV18                                                                                        | <b>EV.5.9 / EV.7.8</b><br>Tractive System connectors outside of a housing must contain an Interlock.                                                                                                                                                       |      |       |         |                                          |                                                                                       |  |    |

|                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                           |             |              |                |                                                                                                   |  |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|----------------|---------------------------------------------------------------------------------------------------|--|-----------|
| <b>4 - EV</b>                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                           | <b>Name</b> | <b>Judge</b> | <b>Judge</b>   | <b>EV Vehicle Number</b>                                                                          |  | <b>4B</b> |
| <b>Check date</b>                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                           |             |              |                |                                                                                                   |  |           |
| <b>EV Tractive System / Car Devices</b>                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                           |             |              |                |                                                                                                   |  |           |
| <b>Item</b>                                                                                                                                                                                    | <b>Description</b>                                                                                                                                                                                                                                                                                                        | <b>Team</b> | <b>Check</b> | <b>Recheck</b> | <b>Recheck (Write on the back if necessary)</b>                                                   |  |           |
| <b>Accumulator Hand Cart</b>                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |             |              |                |                                                                                                   |  |           |
| <b>EV19</b>                                                                                                                                                                                    | <b>EV.4.10.3.a</b><br>The Hand Cart must be able to carry the load of the Accumulator Container(s) without tipping out.                                                                                                                                                                                                   |             |              |                |                                                                                                   |  |           |
| <b>EV20</b>                                                                                                                                                                                    | <b>EV.4.10.3.b</b><br>Hand cart must contain a minimum of two wheels.                                                                                                                                                                                                                                                     |             |              |                |                                                                                                   |  |           |
| <b>EV21</b>                                                                                                                                                                                    | <b>EV.4.10.3.c</b><br>Hand cart must have a brake system using a Dead Man type switch (releasing the brake only when pushing and holding a handle) OR by manually lifting part of the cart off the ground.                                                                                                                |             |              |                |                                                                                                   |  |           |
| <b>EV22</b>                                                                                                                                                                                    | <b>EV.4.10.3.c</b><br>The Hand Cart brake system must stop the hand cart with a fully loaded Accumulator Container.                                                                                                                                                                                                       |             |              |                |                                                                                                   |  |           |
| <b>EV23</b>                                                                                                                                                                                    | <b>EV.4.10.4</b><br>Accumulator Container(s) must be securely attached to the Hand Cart.                                                                                                                                                                                                                                  |             |              |                |                                                                                                   |  |           |
| <b>Accumulator Charger</b>                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                           |             |              |                |                                                                                                   |  |           |
| <b>EV24</b>                                                                                                                                                                                    | <b>EV.8.2.4</b><br>Charger connector must incorporate an interlock such that the connectors only become live if are correctly connected.                                                                                                                                                                                  |             |              |                |                                                                                                   |  |           |
| <b>EV25</b>                                                                                                                                                                                    | <b>EV.8.2.6 / EV.5.8.2 / EV.5.8.3</b><br>The charger must have two TSMPs installed (4mm shrouded red banana jack) and available when charging.<br>TSMPs must be labeled with "HV+" and "HV-".                                                                                                                             |             |              |                |                                                                                                   |  |           |
| <b>EV26</b>                                                                                                                                                                                    | <b>EV.8.2.2</b><br>If the charger housing is conductive, it must be connected to the earth ground of the AC input.                                                                                                                                                                                                        |             |              |                |                                                                                                   |  |           |
| <b>EV27</b>                                                                                                                                                                                    | <b>EV.8.2.3</b><br>All connections of the charger(s) must be isolated and covered.                                                                                                                                                                                                                                        |             |              |                |                                                                                                   |  |           |
| <b>EV28</b>                                                                                                                                                                                    | <b>EV.8.2.5</b><br>High Voltage charging leads must be orange.                                                                                                                                                                                                                                                            |             |              |                |                                                                                                   |  |           |
| <b>EV29</b>                                                                                                                                                                                    | <b>EV.6.3.4.c</b><br>Any shielded cable must have the shield grounded.                                                                                                                                                                                                                                                    |             |              |                |                                                                                                   |  |           |
| <b>EV30</b>                                                                                                                                                                                    | <b>EV.8.2.7</b><br>The charger must include a Push-Pull OR Push-Rotate emergency stop switch. The E-Stop switch must have a minimum diameter of 25mm, open the shutdown circuit and hold the position when pressed.<br>Be labelled with the International Electrical Symbol (a red spark on a white edged blue triangle). |             |              |                | Example:<br> |  |           |
| <b>EV31</b>                                                                                                                                                                                    | <b>EV.6.3.4.a</b><br>Any Tractive System wiring that runs outside of electrical enclosures must be enclosed in separate orange nonconductive conduit OR be a shielded cable.                                                                                                                                              |             |              |                |                                                                                                   |  |           |
| <b>EV32</b>                                                                                                                                                                                    | <b>EV.6.3.4.b</b><br>The conduit or shielded cable must be securely anchored at each end to allow it to withstand a force of 200 N without straining the cable end crimp.                                                                                                                                                 |             |              |                |                                                                                                   |  |           |
| <b>Instruct the ESO to fully gear in PPE and open the accumulator container by removing the lid, and disconnect the Maintenance Plugs. Stop for further checks once the procedure is done.</b> |                                                                                                                                                                                                                                                                                                                           |             |              |                |                                                                                                   |  |           |

|                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|----------------|-------------------------------------------------|--|-----------|
| <b>4 - EV</b>                                                                                                            |                                                                                                                                                                                                                                                                                                                        | <b>Name</b> | <b>Judge</b> | <b>Judge</b>   | <b>EV Vehicle Number</b>                        |  | <b>4C</b> |
| <b>Check date</b>                                                                                                        |                                                                                                                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |
| <b>EV Tractive System / Car Devices</b>                                                                                  |                                                                                                                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |
| <b>Item</b>                                                                                                              | <b>Description</b>                                                                                                                                                                                                                                                                                                     | <b>Team</b> | <b>Check</b> | <b>Recheck</b> | <b>Recheck (Write on the back if necessary)</b> |  |           |
| <b>Accumulator Internal Checks / Container Open</b>                                                                      |                                                                                                                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |
| <b>EV33</b>                                                                                                              | <b>SES CHECK</b><br><b>Carefully check</b> accumulator pictures attached to the inspection sheet and confirm that the accumulator mounts, internal structure and other mechanical parts match what was approved during the SES.                                                                                        |             |              |                |                                                 |  |           |
| <b>EV34</b>                                                                                                              | <b>ESF CHECK</b><br><b>Carefully check</b> all components of the accumulator and car to confirm they match what was provided on the datasheets tab of the ESF. <b>ESF</b> documentation is attached to the inspection sheet.                                                                                           |             |              |                |                                                 |  |           |
| <b>EV35</b>                                                                                                              | <b>EV.5.3.2.b</b><br>Maintenance Plugs must be accessible after opening the Accumulator Container without the need to move/remove any other components.                                                                                                                                                                |             |              |                |                                                 |  |           |
| <b>EV36</b>                                                                                                              | <b>EV.5.3.2.c</b><br>Maintenance Plugs must be impossible to connect in any configuration other than the intended.                                                                                                                                                                                                     |             |              |                |                                                 |  |           |
| <b>EV37</b>                                                                                                              | <b>EV.5.3.2.d</b><br>Maintenance Plugs must not require tools to remove or install. Removal and installation must be possible using HV insulated gloves + leather gloves.                                                                                                                                              |             |              |                |                                                 |  |           |
| <b>EV38</b>                                                                                                              | <b>EV.5.3.2.e</b><br>Maintenance Plugs must include a positive locking feature to prevent unintended disconnection.                                                                                                                                                                                                    |             |              |                |                                                 |  |           |
| <b>EV39</b>                                                                                                              | <b>EV.5.3.2.f</b><br>Maintenance Plugs must have nonconductive surfaces, except where required to make electrical connection.                                                                                                                                                                                          |             |              |                |                                                 |  |           |
| <b>EV40</b>                                                                                                              | <b>EV.7.5.5 / BR.4.6.1.1</b><br>A minimum of 30% of the accumulator cells must have temperature monitoring.                                                                                                                                                                                                            |             |              |                |                                                 |  |           |
| <b>EV41</b>                                                                                                              | <b>EV.7.5.4</b><br>The temperature sensor used must be in direct contact with the cell's negative terminal OR with the busbar less than 10mm away from the cell terminal.                                                                                                                                              |             |              |                |                                                 |  |           |
| <b>EV42</b>                                                                                                              | <b>BR.4.6.1.2</b><br>Attachment of the temperature sensors must not rely solely on tape.                                                                                                                                                                                                                               |             |              |                |                                                 |  |           |
| <b>EV43</b>                                                                                                              | <b>EV.5.2.2.a</b><br>If the container is made of electrically conductive material, the poles of the accumulator stack(s) and cells must be insulated to the inner walls of the Accumulator Container.                                                                                                                  |             |              |                |                                                 |  |           |
| <b>EV44</b>                                                                                                              | <b>EV.6.5.5</b><br>TS and GLV wiring must be separated by an insulating barrier OR distance specified below:<br><ul style="list-style-type: none"> <li>• U &lt; 100 V DC                      10 mm</li> <li>• 100 V DC &lt; U &lt; 200 V DC    20 mm</li> <li>• U &gt; 200 V DC                      30 mm</li> </ul> |             |              |                |                                                 |  |           |
| <b>EV45</b>                                                                                                              | <b>EV.6.6.6</b><br>All wires with more 150mm in length must have a fuse within the first 150mm from the cables's source.                                                                                                                                                                                               |             |              |                |                                                 |  |           |
| <b>EV46</b>                                                                                                              | <b>EV.5.4.3</b><br>No "always energized" TS cables leave the accumulator container (All TS wires leaving accumulator must be disconnected by an AIR).                                                                                                                                                                  |             |              |                |                                                 |  |           |
| <b>Spare parts</b>                                                                                                       |                                                                                                                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |
| <b>EV47</b>                                                                                                              | <b>EV.5.1.4</b><br>All Accumulator Segments and/or Accumulator Containers (including spares and replacement parts) must be identical to the design documented in the ESF and SES.                                                                                                                                      |             |              |                |                                                 |  |           |
| <b>Instruct the ESO (while still fully geared in PPE) to close the accumuator container and bolt the lid into place.</b> |                                                                                                                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |

|                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                             |             |              |                |                                                 |  |           |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|----------------|-------------------------------------------------|--|-----------|
| <b>4 - EV</b>                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                             | <b>Name</b> | <b>Judge</b> | <b>Judge</b>   | <b>EV Vehicle Number</b>                        |  | <b>4D</b> |
| <b>Check date</b>                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                             |             |              |                |                                                 |  |           |
| <b>EV Tractive System / Car Devices</b>                                                                              |                                                                                                                                                                                                                                                                                                                                                                             |             |              |                |                                                 |  |           |
| <b>Item</b>                                                                                                          | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                          | <b>Team</b> | <b>Check</b> | <b>Recheck</b> | <b>Recheck (Write on the back if necessary)</b> |  |           |
| <div style="text-align: center;"> <b>Charging Procedure - CAUTION: ENERGIZED</b> </div>                              |                                                                                                                                                                                                                                                                                                                                                                             |             |              |                |                                                 |  |           |
| <b>EV48</b>                                                                                                          | <b>EV.11.5.4</b><br>When charging, each Accumulator Container(s) must have a label that contains:<br><ul style="list-style-type: none"> <li>Team Name</li> <li>Electrical System Officer (ESO) phone number(s)</li> </ul>                                                                                                                                                   |             |              |                |                                                 |  |           |
| <b>EV49</b>                                                                                                          | <b>EV.6.7</b><br>Measured resistance between conductive surfaces and ground pin of the charger's AC connector must be < 300mOhm.                                                                                                                                                                                                                                            |             |              |                |                                                 |  |           |
| <b>EV50</b>                                                                                                          | <b>EV.5.8.2 / EV.8.2.6</b><br>Measure resistance between TSMP HV- and TSMP HV+:<br><ul style="list-style-type: none"> <li>Vmax &lt; 200V = 10kOhm</li> <li>200V &lt; Vmax &lt; 400V = 20kOhm</li> <li>400V &lt; Vmax &lt; 600V = 30kOhm</li> </ul>                                                                                                                          |             |              |                |                                                 |  |           |
| <b>EV51</b>                                                                                                          | <b>EV.5.7</b><br>Accumulator Indicator indicates if voltage above 60VDC is present outside of the container.<br>Check indicator during the charging demonstration.                                                                                                                                                                                                          |             |              |                |                                                 |  |           |
| <b>EV52</b>                                                                                                          | <b>EV.5.7.4.a</b><br>The Voltage Indicator must be clearly visible when connecting/disconnecting the Accumulator Tractive System connections.                                                                                                                                                                                                                               |             |              |                |                                                 |  |           |
| <b>EV53</b>                                                                                                          | <b>EV.8.2.7 / EV.8.4.2</b><br>When charging, press the E-Stop button.<br>The E-Stop button on charger must open the shutdown circuit and stop the charging.                                                                                                                                                                                                                 |             |              |                |                                                 |  |           |
| <b>EV54</b>                                                                                                          | <b>EV.8.3 / EV.8.4.1 / IN.4.4</b><br>IMD must open shutdown circuit in 30 seconds when isolation fault is present.<br>1. Accumulator in valid charge condition.<br>2. Connect TSMP (-) and GLVS ground using the IMD test resistor.<br>3. IMD shall detect an insulation fault, turn off the charger and open AIRs in 30 seconds.<br>4. Repeat the test using the TSMP (+). |             |              |                |                                                 |  |           |
| <b>EV55</b>                                                                                                          | <b>EV.7.3 / EV.7.4.1 / EV.7.5.5 / BR.4.6.1.1</b><br>When charging, the BMS must be live, monitoring the cell voltage of each cell and cell temperature of more than 30% of cells.                                                                                                                                                                                           |             |              |                |                                                 |  |           |
| <b>EV56</b>                                                                                                          | <b>EV.7.3.5 / EV.8.3</b><br>When charging, the BMS must be able to turn OFF the charger in a event that a fault is detected (Shutdown Circuit open).                                                                                                                                                                                                                        |             |              |                |                                                 |  |           |
| <b>EV57</b>                                                                                                          | <b>IN.2.6 / IN.4.7.1</b><br>Upon completion of accumulator inspection the following parts must be sealed to prevent modification:<br><ul style="list-style-type: none"> <li>Accumulator Container openings.</li> <li>HV Main Connector.</li> <li>Charger.</li> </ul>                                                                                                        |             |              |                |                                                 |  |           |
| <b>Accumulator technical inspection is complete, contact the Tech. Inspection lead to acquire accumulator seals.</b> |                                                                                                                                                                                                                                                                                                                                                                             |             |              |                |                                                 |  |           |

|                                  |                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
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| 4 - EV                           |                                                                                                                                                                                                                                                                                                                                                         | Name | Judge | Judge   | EV Vehicle Number                        |  | 4E |
|                                  |                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| Check date                       |                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| EV Tractive System / Car Devices |                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| Item                             | Description                                                                                                                                                                                                                                                                                                                                             | Team | Check | Recheck | Recheck (Write on the back if necessary) |  |    |
| EV Mechanical Inspection         |                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| EV58                             | <b>IN.2.3</b><br>Visible access must be possible to all items on Tech Sheet without the use of mirrors, borescopes, etc.                                                                                                                                                                                                                                |      |       |         |                                          |  |    |
| EV59                             | <b>T.1.9</b><br>The cockpit firewall must separate the driver from any Tractive System components other than Outboard Wheel Motors and associated cables. Must be made of aluminum and grounded to the chassis.                                                                                                                                         |      |       |         |                                          |  |    |
| EV60                             | <b>FOOT PEDAL ACCELERATOR</b><br>The foot pedal must have a positive stop to prevent sensors from being mechanically overstressed.Two springs must be used to return the accelerator pedal to the off position and each spring must work with the other disconnected. The springs in the APPS are not acceptable return springs.                        |      |       |         |                                          |  |    |
| EV61                             | <b>T.3.3</b> <ul style="list-style-type: none"><li>• Brake pedal overtravel switch must open shutdown circuit.</li><li>• Not resetable, must not rely on software to work.</li></ul>                                                                                                                                                                    |      |       |         |                                          |  |    |
| EV62                             | <b>IN.2.6 / IN.4.7.1</b><br>LV Batteries and Pressurized cylinders must be: <ul style="list-style-type: none"><li>•Inside F.1.1 chassis</li><li>•Below Shoulder Harness.</li><li>•Below top of chassis ahead &amp; behind &amp; to the left &amp; to the right.</li></ul>                                                                               |      |       |         |                                          |  |    |
| EV63                             | <b>T.6.2</b><br>High pressure hydraulic pumps and lines must have 1 mm thick steel or aluminum shields to protect driver and workers. (Brakes & clutch exempt.)                                                                                                                                                                                         |      |       |         |                                          |  |    |
| EV64                             | <b>T.5.6</b><br>Catch cans must: <ul style="list-style-type: none"><li>•Must have 0.9 L of min vol. •Oil(s) and water(s) must be separate.</li><li>•100 °C-capable material. •Behind firewall, below shoulder level. •3 mm min diameter vent, directed away from driver. •Trans, diff, other systems (unless sealed): 10% or 0.5 L catch can.</li></ul> |      |       |         |                                          |  |    |
| EV65                             | <b>T.5.5.1</b><br>Fluid Leaks are not permitted. Firewalls must prevent contact w/ driver.                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
| EV67                             | <b>T.5.2</b><br>Scattershields are required for clutches, chains, belts, CVT rotating parts, etc. Must not be perforated and have to end parallel to lowest part of front and rear sprockets. Attached with min. 6mm fasteners (CRITICAL FASTENERS)                                                                                                     |      |       |         |                                          |  |    |
| EV68                             | <b>T.5.2</b><br>Scattershields must be made of:<br>For chains: 2.7 mm (0.105”) min thick steel, 3x chain width;<br>For belts: 3 mm (0.12”) min thick aluminum 6061-T6, 1.7x belt width.<br>OEM motor drive sprocket cover is OK.                                                                                                                        |      |       |         |                                          |  |    |
| EV69                             | <b>T.5.2.10</b><br>Finger guards are required to cover all drivetrain parts that spin while car is at rest. No holes >12 mm dia.                                                                                                                                                                                                                        |      |       |         |                                          |  |    |
| EV70                             | <b>T.6</b><br>Compressed gas cylinders must:<br>Be an unmodified COTS cylinder (labeled), filled with nonflammable gas and with a regulator on tank.<br>Securely mounted, axis not pointed at driver, rearward of Main Hoop within the frame envelope, or in structural sidepod; not in cockpit, with appropriate lines & fittings.                     |      |       |         |                                          |  |    |
| EV71                             | <b>T.5.4</b><br>Coolants must be 100% water, or oil.                                                                                                                                                                                                                                                                                                    |      |       |         |                                          |  |    |

| 4 - EV                           |                                                                                                                                                                                                                                                                                                                                              | Name | Judge | Judge   | EV Vehicle Number                        |                                                                                       | 4F |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------|------------------------------------------|---------------------------------------------------------------------------------------|----|
| Check date                       |                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |                                                                                       |    |
| EV Tractive System / Car Devices |                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |                                                                                       |    |
| Item                             | Description                                                                                                                                                                                                                                                                                                                                  | Team | Check | Recheck | Recheck (Write on the back if necessary) |                                                                                       |    |
| TS Design and Components         |                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |                                                                                       |    |
| EV72                             | <b>F.11.1 / F.5.16 / F.3.2.1.n / F.4.1.3</b><br>TS components (except only for Outboard Wheel Motors and its connections) must be contained inside the Rollover Protection Envelope.<br>                                                                    |      |       |         |                                          |                                                                                       |    |
| EV73                             | <b>EV.6.3.2</b><br>TS wiring must be marked with wire gauge, temperature rating ( $\geq 90^{\circ}\text{C}$ ) and insulation voltage rating.                                                                                                                                                                                                 |      |       |         |                                          |                                                                                       |    |
| EV74                             | <b>EV.6.3.3</b><br>TS wiring must be protected from loosening due to vibration, against damage by rotating and/or moving parts, and located out of the way of possible snagging or damage.                                                                                                                                                   |      |       |         |                                          |                                                                                       |    |
| EV75                             | <b>EV.6.3.4</b><br>TS wiring must be enclosed in separate orange nonconductive conduit OR be an orange shielded cable.<br>The conduit or cable must be securely anchored at each end to allow it to withstand a force of 200N without straining the cable or crimp.                                                                          |      |       |         |                                          |                                                                                       |    |
| EV76                             | <b>EV.6.7</b><br>Measure resistance between electrically conductive surfaces and GLV System Ground must be $< 300\text{m}\Omega$ .<br>Measure resistance between Parts which may become electrically conductive (Carbon Fiber parts) and GLV System Ground must be $< 50\text{m}\Omega$ .                                                    |      |       |         |                                          |                                                                                       |    |
| EV77                             | <b>EV.4.9.1</b><br>Each housing or enclosure containing parts of the Tractive System other than Motor housings, must be labelled with:<br>• Symbol specified in ISO 7010-W012 (triangle with black lightning bolt on yellow background)<br>• Text "High Voltage" if the voltage $> 60\text{VDC}$                                             |      |       |         | Example:                                 |  |    |
| EV78                             | <b>EV.5.8.1</b><br>TSMPs must be:<br>• Next to the Master Switches<br>• Protected by a nonconductive housing that can be opened without tools<br>• Protected from being touched with bare hands/fingers once the housing is opened                                                                                                           |      |       |         |                                          |                                                                                       |    |
| EV79                             | <b>EV.5.8.3</b><br>TSMPs must be 4mm shrouded red banana jacks, marked with "HV-" and "HV+".                                                                                                                                                                                                                                                 |      |       |         |                                          |                                                                                       |    |
| EV80                             | <b>EV.5.8.2</b><br>Measure resistance between TSMP HV- and TSMP HV+:<br>• $000\text{V} < V_{\text{max}} < 200\text{V} = 10\text{k}\Omega + \text{discharge}$<br>• $200\text{V} < V_{\text{max}} < 400\text{V} = 20\text{k}\Omega + \text{discharge}$<br>• $400\text{V} < V_{\text{max}} < 600\text{V} = 30\text{k}\Omega + \text{discharge}$ |      |       |         |                                          |                                                                                       |    |
| EV81                             | <b>EV.6.1.2</b><br>When the Tractive System enclosures are in place, must not be possible to contact any Tractive System connections with a 100mm long, 6mm diameter insulated test probe.                                                                                                                                                   |      |       |         |                                          |                                                                                       |    |
| EV82                             | <b>EV.6.2.1</b><br>Insulation materials must have a minimum temperature rating of $90^{\circ}$ .                                                                                                                                                                                                                                             |      |       |         |                                          |                                                                                       |    |
| EV83                             | <b>EV6.2.2</b><br>Insulating tape or paint may be part of the insulation but must not be the only insulation.                                                                                                                                                                                                                                |      |       |         |                                          |                                                                                       |    |
| EV84                             | <b>EV.6.4.3 / EV.6.4.4</b><br>Bolted electrical connections in the high current path of the Tractive System must include a positive locking feature to prevent unintentional loosening.                                                                                                                                                      |      |       |         |                                          |                                                                                       |    |



|                                         |                                                                                                                                                                                                                                                                           |             |              |                |                                                 |                                                                                       |           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|----------------|-------------------------------------------------|---------------------------------------------------------------------------------------|-----------|
| <b>4 - EV</b>                           |                                                                                                                                                                                                                                                                           | <b>Name</b> | <b>Judge</b> | <b>Judge</b>   | <b>EV Vehicle Number</b>                        |                                                                                       | <b>4G</b> |
| <b>Check date</b>                       |                                                                                                                                                                                                                                                                           |             |              |                |                                                 |                                                                                       |           |
| <b>EV Tractive System / Car Devices</b> |                                                                                                                                                                                                                                                                           |             |              |                |                                                 |                                                                                       |           |
| <b>Item</b>                             | <b>Description</b>                                                                                                                                                                                                                                                        | <b>Team</b> | <b>Check</b> | <b>Recheck</b> | <b>Recheck (Write on the back if necessary)</b> |                                                                                       |           |
| <b>EV85</b>                             | <b>EV.7.8 / EV.4.1.3 / EV.5.5.2 / EV.5.9</b><br>Interlocks must be incorporated into:<br>• Tractive System Connectors<br>• MSD<br>• Outboard Wheel Motors                                                                                                                 |             |              |                |                                                 |                                                                                       |           |
| <b>EV86</b>                             | <b>EV.5.5.1.b</b><br>MSD must be more than 350mm from the ground.                                                                                                                                                                                                         |             |              |                |                                                 |                                                                                       |           |
| <b>EV87</b>                             | <b>EV.5.5.1.c</b><br>MSD must be easily visible when standing behind the vehicle.                                                                                                                                                                                         |             |              |                |                                                 |                                                                                       |           |
| <b>EV88</b>                             | <b>EV.5.5.1.d</b><br>MSD must be operable in 10 seconds or less by an untrained person.                                                                                                                                                                                   |             |              |                |                                                 |                                                                                       |           |
| <b>EV89</b>                             | <b>EV.5.5.1.e</b><br>MSD must be operable without removing any bodywork or obstruction or using tools.                                                                                                                                                                    |             |              |                |                                                 |                                                                                       |           |
| <b>EV90</b>                             | <b>EV.5.5.1.g</b><br>MSD must be clearly marked with "MSD".                                                                                                                                                                                                               |             |              |                |                                                 |                                                                                       |           |
| <b>EV91</b>                             | <b>EV.5.5.4 / EV.6.1.2</b><br>A dummy connector or similar may be used on the MSD to restore isolation.                                                                                                                                                                   |             |              |                |                                                 |                                                                                       |           |
| <b>EV92</b>                             | <b>EV.7.10.2</b><br>Shutdown buttons must be a push-pull OR push-rotate emergency stop button. Must hold when operated to the OFF position.                                                                                                                               |             |              |                |                                                 |                                                                                       |           |
| <b>EV93</b>                             | <b>EV.7.10.3</b><br>One Shutdown button must be on each side of the vehicle. It must have a diameter 40mm minimum. Must not be easily removable or mounted onto removable bodywork.                                                                                       |             |              |                |                                                 |                                                                                       |           |
| <b>EV94</b>                             | <b>EV.7.10.4</b><br>One Shutdown button must be mounted in the cockpit. It must have a diameter 24mm minimum.<br>Located in easy reach of the belted in driver and not obstructed by the steering wheel or any other part.                                                |             |              |                |                                                 |                                                                                       |           |
| <b>EV95</b>                             | <b>EV.7.10.5</b><br>The Shutdown buttons must have the International Electrical symbol.<br>(a red spark on a white edged blue triangle)                                                                                                                                   |             |              |                | Example:                                        |  |           |
| <b>EV96</b>                             | <b>EV.7.9 / T.9.3</b><br>Each vehicle must have two Master Switches located on driver's right-hand side, at the driver's shoulder height and in proximity to the Main Hoop.<br>OFF and ON position must be clearly marked.<br>ON position must be in horizontal position. |             |              |                |                                                 |                                                                                       |           |
| <b>EV97</b>                             | <b>EV.7.9.2</b><br>GLVMS must be in the center of a completely red circular area of >50mm in diameter. Must be labeled "LV".                                                                                                                                              |             |              |                |                                                 |                                                                                       |           |
| <b>EV98</b>                             | <b>EV.7.9.3</b><br>TSMS must be in the center of a completely orange circular area of >50mm in diameter.<br>Must be labeled "TS" and the symbol specified in ISO 7010-W012.<br>(triangle with black lightning bolt on yellow background)                                  |             |              |                | Example:                                        |  |           |
| <b>EV99</b>                             | <b>EV.7.9.3.e / EV.11.3.1</b><br>Master switches must be fitted with a "Lockout/Tagout" capability in the OFF position.                                                                                                                                                   |             |              |                |                                                 |                                                                                       |           |
| <b>EV100</b>                            | <b>IN.4.5</b><br>Measure isolation between TSMP and chassis ground.<br>Choose next voltage level above TS Voltage (250V or 500V)<br>$R_{iso} \geq 500 * TS \text{ Voltage} + BPR$                                                                                         |             |              |                |                                                 |                                                                                       |           |

4 - EV

|            |  |
|------------|--|
| Check date |  |
|------------|--|

EV Tractive System / Car Devices

| 4 - EV                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Name | Judge | Judge   | EV Vehicle Number                        |  | 4H |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------|------------------------------------------|--|----|
| Check date                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |       |         |                                          |  |    |
| EV Tractive System / Car Devices                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |       |         |                                          |  |    |
| Item                                                                                                                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Team | Check | Recheck | Recheck (Write on the back if necessary) |  |    |
| <div><div></div><div>CAUTION: ENERGIZED</div><div></div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |       |         |                                          |  |    |
| EV101                                                                                                                                                                                                                              | <b>EV.7.6 / EV.5.11</b><br>IMD must open the shutdown circuit in 30 seconds when isolation fault is present.<br><b>1.</b> Activate Tractive System. IMD indicator OFF, TSSI is GREEN.<br><b>2.</b> Connect IMD test box between TSMP HV- and GLV Ground.<br><b>3.</b> IMD must trip in 30 seconds and TS voltage must decrease below 60VDC in 5 seconds. IMD indicator is ON (RED). TSSI is flashing RED.<br><b>4.</b> Remove IMD test box and wait 30 seconds. DOES NOT RESET THE FAULT. TS stays OFF. IMD indicator stays ON. TSSI stays flashing RED.<br><b>5.</b> Reset the fault and repeat the test procedure with TSMP HV+. |      |       |         |                                          |  |    |
| EV102                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, turn the <b>TSMS OFF</b> .<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |       |         |                                          |  |    |
| EV103                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, turn the <b>GLVMS OFF</b> .<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |       |         |                                          |  |    |
| EV104                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, push the <b>Left Shutdown Button</b> .<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |         |                                          |  |    |
| EV105                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, push the <b>Cockpit Shutdown Button</b> .<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |         |                                          |  |    |
| EV106                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, turn the <b>BOTS OFF</b> .<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |       |         |                                          |  |    |
| EV107                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, remove the <b>MSD OFF</b> .<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |       |         |                                          |  |    |
| EV108                                                                                                                                                                                                                              | <b>EV.7.2.2</b><br>When all shutdown circuit switches are ON, open the <b>Outboard Wheel Motor Interlocks</b> (if applicable).<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |       |         |                                          |  |    |
| EV109                                                                                                                                                                                                                              | <b>T.9.4 / EV.7.2.2</b><br>Unmount the Inertia Switch. Activate TS, shake the switch and check if TS deactivates.<br>TS Voltage must decrease below 60VDC in 5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |       |         |                                          |  |    |
| EV110                                                                                                                                                                                                                              | <b>EV.5.10</b><br>The RTML must be flashing amber when GLV turned ON and the voltage outside the accumulator container exceeds 60VDC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |         |                                          |  |    |
| EV111                                                                                                                                                                                                                              | <b>EV.5.10</b><br>The RTML must comply the DOT FMVSS 108.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |         |                                          |  |    |
| EV112                                                                                                                                                                                                                              | <b>EV.9.6</b><br>The vehicle must make a characteristic sound when its Ready-to-Drive.<br>Must be sounded continuously for min 1 second and max 3 seconds. Minimum sound level of 80dBA.<br>The vehicle must not make other sounds similar to Ready-to-Drive sound.                                                                                                                                                                                                                                                                                                                                                                |      |       |         |                                          |  |    |
| EV113                                                                                                                                                                                                                              | <b>T.3.4.1</b><br>Press Brake Pedal.<br>The Brake Light must be clearly visible from the rear in very bright sunlight.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |       |         |                                          |  |    |
| EV114                                                                                                                                                                                                                              | <b>T.3.4.2</b><br>Brake Light must be Red in color on a Black background with a minimum shining surface of 15cm².                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |       |         |                                          |  |    |

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |    |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------|------------------------------------------|----|
| <b>4 - EV</b>                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Name | Judge | Judge   | EV Vehicle Number                        | 41 |
| Check date                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |    |
| <b>EV Tractive System / Car Devices</b>                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |    |
| Item                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               | Team | Check | Recheck | Recheck (Write on the back if necessary) |    |
| <div style="text-align: center;"> <b>CAUTION: ENERGIZED</b> </div>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |    |
| EV115                                                                                                | <b>EV.9.5</b><br>The team must demonstrate how the car is set to Ready-to-Drive mode by the driver.<br>1. Enable TS<br>2. Press Accelerator, motors SHOULD NOT spin<br>3. Press brake and start button<br>4. Press Accelerator, motors SHOULD spin<br>5. Press shutdown button<br>6. Release shutdown button and enable TS<br>7. Press Accelerator, motors SHOULD NOT spin<br>8. Press brake and start button<br>9. Press Accelerator, motors SHOULD spin |      |       |         |                                          |    |
| EV116                                                                                                | <b>EV.4.7</b><br>Torque production must stop when accelerator and brake pedal are pressed simultaneously.<br>1. Press accelerator to show that axle turns.<br>2. With accelerator > 25%, press brake pedal. Axle must stop.<br>3. Keeping accelerator >25%, release brake pedal. Axle must remain stopped.<br>4. Slowly release accelerator, axle may turn again once < 5% pedal position.                                                                |      |       |         |                                          |    |
| EV117                                                                                                | <b>EV.4.2.9</b><br>Torque production must stop if APPS are implausible.<br>1. Press accelerator to show that axle turns.<br>2. Unplug all but 1 APPS.<br>3. Press accelerator, axle must not turn.<br>Repeat for each APPS sensor.                                                                                                                                                                                                                        |      |       |         |                                          |    |
| EV118                                                                                                | <b>EV.7.7</b><br>The team must have a test to demonstrate the BSPD operation.<br>1. Enable TS<br>2. Press and hold Brake (may require driver in car)<br>3. Turn up simulated current (5kW at nominal voltage)<br>4. TS must shut off in 0.5 sec<br>5. Release brake and turn off simulated current. The TS may not automatically return to active state. The driver must not be able to reactive the Tractive System.                                     |      |       |         |                                          |    |
| EV119                                                                                                | <b>EV.3.2</b><br>The Energy Meter must be installed in an easily accessible location.<br><br>Energy Meter code:                                                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |    |
| EV120                                                                                                | <b>IN.2.6 / IN.4.7.1</b><br>Uppon completion of the EV inspection the following parts must be sealed to prevent modification: <ul style="list-style-type: none"> <li>• Energy Meter housing.</li> <li>• Any TS housing openings.</li> </ul>                                                                                                                                                                                                               |      |       |         |                                          |    |
| EV technical inspection is complete, contact the Tech. Inspection lead to acquire accumulator seals. |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |         |                                          |    |

## EGRESS TEST (5 seconds) 2025

|               |  |                   |                 |  |             |
|---------------|--|-------------------|-----------------|--|-------------|
| <b>CAR #</b>  |  | <b>Team:</b>      |                 |  | <b>Date</b> |
| <b>Judge:</b> |  | <b>Start Time</b> | <b>End Time</b> |  |             |

### Instructions

#### THE CAR MUST HAVE THE MECHANICAL TECHNICAL INSPECTION STICKER

- 1) All side and front body parts must be mounted.
- 2) Drivers must wear all equipment (including arm restraints).
- 3) Helmet must be 50mm (2") from the line between the top of the Front Hoop and Main Hoop, and from the line between the top of the Main Hoop and the base of the Main Hoop Bracing. The driver must be able to steer the wheel without issues.
- 4) Lap Belt - Above the hip bones and thighs.
- 5) Shoulder Harness - Between 10° above and 20° below horizontal, and properly tightened.
- 6) Anti-Submarine Straps - Properly tightened.
- 7) Less than 5 seconds between the driver removing one hand from the steering wheel and placing BOTH feet on the ground in a ready-to-run condition.
- 8) During exit, the Kill Switch must be activated.

\* The tallest approved driver must also receive the tallest driver wristband (ensure the wristband is placed on the correct person; do not simply hand it to the team).

\* If a taller driver appears after the car has been approved on the Tilt-Table, the team will lose the Tilt-Table sticker (IN.9.1b, IN.13.3.1).

\* A maximum of 2 drivers at a time will be allowed to perform the 5-second exit with the Tilt-Table sticker. Subsequent drivers will go to the end of the line. If the team does not have the Tilt-Table sticker, they may have as many drivers as desired perform the exit at once.

In case of conflict with this document, the FSAE Rules 2025 v1 prevail.

### ATTENTION DRIVERS:

**By signing this form, you attest that you have watched the dynamic event instruction video and acknowledge its content.**

|          |                |                                 |                         |                          |
|----------|----------------|---------------------------------|-------------------------|--------------------------|
| <b>1</b> | Bracelet Code: | Name:                           | <b>Egress Time (s):</b> | <b>Driver Signature:</b> |
|          | Height (m):    | Helmet 50mm (2"): clearance OK? |                         |                          |
| <b>2</b> | Bracelet Code: | Name:                           | <b>Egress Time (s):</b> | <b>Driver Signature:</b> |
|          | Height (m):    | Helmet 50mm (2"): clearance OK? |                         |                          |
| <b>3</b> | Bracelet Code: | Name:                           | <b>Egress Time (s):</b> | <b>Driver Signature:</b> |
|          | Height (m):    | Helmet 50mm (2"): clearance OK? |                         |                          |
| <b>4</b> | Bracelet Code: | Name:                           | <b>Egress Time (s):</b> | <b>Driver Signature:</b> |
|          | Height (m):    | Helmet 50mm (2"): clearance OK? |                         |                          |
| <b>5</b> | Bracelet Code: | Name:                           | <b>Egress Time (s):</b> | <b>Driver Signature:</b> |
|          | Height (m):    | Helmet 50mm (2"): clearance OK? |                         |                          |
| <b>6</b> | Bracelet Code: | Name:                           | <b>Egress Time (s):</b> | <b>Driver Signature:</b> |
|          | Height (m):    | Helmet 50mm (2"): clearance OK? |                         |                          |

## EGRESS TEST (5 seconds) 2025 - Continued

|        |  |            |          |  |      |
|--------|--|------------|----------|--|------|
| CAR #  |  | Team:      |          |  | Date |
| Judge: |  | Start Time | End Time |  |      |

|   |                |                                    |                  |                   |
|---|----------------|------------------------------------|------------------|-------------------|
| 7 | Bracelet Code: | Name:                              | Egress Time (s): | Driver Signature: |
|   | Height (m):    | Helmet 50mm (2"):<br>clearance OK? |                  |                   |

|   |                |                                    |                  |                   |
|---|----------------|------------------------------------|------------------|-------------------|
| 8 | Bracelet Code: | Name:                              | Egress Time (s): | Driver Signature: |
|   | Height (m):    | Helmet 50mm (2"):<br>clearance OK? |                  |                   |

|   |                |                                    |                  |                   |
|---|----------------|------------------------------------|------------------|-------------------|
| 9 | Bracelet Code: | Name:                              | Egress Time (s): | Driver Signature: |
|   | Height (m):    | Helmet 50mm (2"):<br>clearance OK? |                  |                   |

|    |                |                                    |                  |                   |
|----|----------------|------------------------------------|------------------|-------------------|
| 10 | Bracelet Code: | Name:                              | Egress Time (s): | Driver Signature: |
|    | Height (m):    | Helmet 50mm (2"):<br>clearance OK? |                  |                   |

|    |                |                                    |                  |                   |
|----|----------------|------------------------------------|------------------|-------------------|
| 11 | Bracelet Code: | Name:                              | Egress Time (s): | Driver Signature: |
|    | Height (m):    | Helmet 50mm (2"):<br>clearance OK? |                  |                   |

|    |                |                                    |                  |                   |
|----|----------------|------------------------------------|------------------|-------------------|
| 12 | Bracelet Code: | Name:                              | Egress Time (s): | Driver Signature: |
|    | Height (m):    | Helmet 50mm (2"):<br>clearance OK? |                  |                   |

NON-CONFORMITY COMMENTS:

|  |
|--|
|  |
|--|

## RAIN TEST 2025

|                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                     |             |           |  |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|--|------|
| CAR #                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                     | Team:       |           |  | Date |
| Judge:                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                     | Start time: | End Time: |  |      |
| <b>Instructions:</b>                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                     |             |           |  |      |
| <b>THE CAR MUST HAVE THE STICKER FOR APPROVAL IN THE MECHANICAL / EV INSPECTION</b><br>1) The car must be complete (ready to Drive);<br>2) All accompanying members must be using PPE;<br>3) The vehicle must be off the ground in its jacking points;<br>4) No driver/member is inside the vehicle;<br>5) The vehicle must be completely off. |                                                                                                                                                                                                                                                                     |             |           |  |      |
| <b>Procedure:</b>                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                     |             |           |  |      |
| 1                                                                                                                                                                                                                                                                                                                                              | Instruct the ESO to activate the TS, wait for the RML to flash;                                                                                                                                                                                                     |             |           |  |      |
| 2                                                                                                                                                                                                                                                                                                                                              | Inspect the TSSI for the IMD, confirming that there are no insulation faults. Then, instruct the ESO to connect a resistor to the TSMP, inducing a fault.<br>Inspect the TSSI for the IMD again, confirming that the system is operational and detecting the fault. |             |           |  |      |
| 3                                                                                                                                                                                                                                                                                                                                              | Reset the system and activate the TS again.                                                                                                                                                                                                                         |             |           |  |      |
| 4                                                                                                                                                                                                                                                                                                                                              | With the vehicle on jack stands and no personnel nearby, start the water spray.<br>Keep the spray on for 120 seconds.                                                                                                                                               |             |           |  |      |
| 5                                                                                                                                                                                                                                                                                                                                              | Stop the water spray and monitor the vehicle for another 120 seconds.                                                                                                                                                                                               |             |           |  |      |
| 6                                                                                                                                                                                                                                                                                                                                              | The test is considered successful if the vehicle does not enter a fault condition for the entire 240 seconds.                                                                                                                                                       |             |           |  |      |

**If the test is successful, it is considered complete. Contact the EV inspection leader to obtain the approval sticker.**

**NON-CONFORMITY COMMENTS:**

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